Balanced circle packings from terminal tangle decorations

2 November 2026

Abstract

A matched triangulation of the sphere specifies an augmented link and a complete circle packing. We give a finite modification near its crossing disks that makes a strict shared-face inequality hold at every crossing region. The modification leaves every original circle fixed and inserts an even number of terminal tangles on each side of each original region. The two insertion counts are chosen by an explicit formula. Their geometric recurrence and an integral for the normalized triangle action prove strict balance for every retained and added region, including regions adjacent to the two marked lines. Odd powers in the added tangles preserve the original endpoint pairing. The shared-face criterion then gives the volume limit for the resulting fixed knots after independently sufficiently long signed fillings. The construction can change the knot; all powers and all decorations are fixed before the level tends to infinity.

1 The diagram, the inequality, and the result

A *marked matched sphere* consists of a finite oriented simple triangulation of the two-sphere, a perfect matching of its dual graph, and a chosen edge outside that matching. A dual matching edge pairs two adjacent triangles. We also call their common primal edge a matching edge. Every triangle belongs to exactly one pair. The chosen nonmatching edge is the *mark*. Simplicity means that edges have distinct endpoints, no two vertices are joined by more than one edge, and every face is a triangle with three distinct vertices. There are at least four vertices.

The inverse projection and crossing-disk cut places a two-strand crossing-circle region at each matched pair and joins its four terminals along the remaining dual edges. It gives an augmented link in S^3. A crossing circle bounds the specified disk meeting the main strands twice. The cyclic order and the frame of that disk are part of the data. We retain either no crossing or a signed half twist $\epsilon_e \in \{-1, 0, 1\}$ at each region e. Filling the crossing circle along its disk-framed primitive slope $\mu_e + p_e\lambda_e$ produces the physical power

$$n_e = \epsilon_e - 2p_e. \tag{1.1}$$

Its parity is fixed by ϵ_e. We require that the chosen parities join the main strands into one knot, with no additional closed main component.

The complete packing can be normalized by sending the contact at the mark to infinity. Its two incident white circles become parallel lines. Their curvatures are zero; every other white circle has positive curvature w_f, the reciprocal of its Euclidean radius. We call these remaining vertices ordinary. All occurrences of a vertex refer to the same circle and the same weight.

The action used below has a fixed additive normalization. Define

$$\Lambda(t) = -\int_0^t \log(2\sin s)\, ds, \qquad e(a) = \frac{\Lambda(\pi a) + \Lambda(\pi/2 - \pi a)}{\pi}, \quad 0 \le a \le \tfrac{1}{2},$$

and, for nonnegative weights with at least one positive entry, put

$$\mathcal{Q}(w_1,w_2,w_3)=\max_{\substack{a_i\ge 0\\ \sum_i a_i=1/2\\ a_i=0 \text{ if } w_i=0}}\left\{\sum_i e(a_i)+\sum_{w_i>0} a_i\log w_i\right\}. \tag{1.2}$$

In particular $\mathcal{Q}(0,w,0)=\frac{1}{2}\log w$. This is the triangle action in the shared-face coefficient estimate; we do not add a constant depending on the triangle.

At a matched pair the two oriented triangles are (x,b,y) and (y,c,x). Its cyclic boundary tuple is (x,b,y,c). We write

$$H(x,b,c,y) \tag{1.3}$$

for a local source polynomial: its outer variables are x,y, and its input and output middle variables are b,c. Thus the argument order in (1.3) differs from the cyclic boundary order. The same letters will denote the corresponding curvatures when no polynomial is being evaluated. The matching edge xy cannot be the marked nonmatching edge, so

$$A=x+y>0,\qquad P=xy,\qquad \rho=P/A^2\in[0,1/4]. \tag{1.4}$$

Definition 1.1 (Strict complete balance)**.** A region with weights (x,b,c,y) is strictly balanced if

$$X:=\mathcal{Q}(x,b,y)+\mathcal{Q}(x,c,y)>\log b\quad(b>0),\qquad X>\log c\quad(c>0).$$

A comparison with a zero middle is omitted.

Theorem 1.2 (Finite balanced decoration)**.** *Fix a marked matched sphere with its crossing-disk frames and a knot parity pattern. There is a finite decoration by terminal crossing-circle tangles with the following properties.*

(i) *At each original region an even nonnegative number of terminals is inserted on each of its two shaded sides. No original crossing circle is removed.*

(ii) *The refined white sphere is simple and has its original nonmatching marked edge. Its complete packing extends the original packing, leaving every old circle fixed.*

(iii) *Every original and added crossing region is strictly balanced at this one complete packing.*

(iv) *Give every added terminal an odd power, of either sign. The four-endpoint pairing at each decorated original region is unchanged, and no internal main component is created.*

The insertion counts are the explicit integers in (5.3). The construction uses no bound on the number of regions and no acyclicity assumption on the original diagram.

Corollary 1.3 (Filled knot volume)**.** *Let $\widehat{D}$ be a fixed decorated diagram from Theorem 1.2. Retain the original knot parity pattern and odd parity at every added terminal. There is a threshold $T_{\widehat{D}}$ such that, for every fixed collection of integers p_e with $|p_e|\ge T_{\widehat{D}}$, independently in sign and size, the resulting knot $K_{\boldsymbol{p}}$ is hyperbolic and*

$$\lim_{N\to\infty}\frac{2\pi}{N}\log|\langle K_{\boldsymbol{p}}\rangle_N|=\mathrm{Vol}(S^3\setminus K_{\boldsymbol{p}}). \tag{1.5}$$

Here $\langle K\rangle_N$ is the reduced zero-framed Kashaev invariant, with unknot value one and parameters $q_N=e^{\pi\mathrm{i}/N}$, $q_N^2=e^{2\pi\mathrm{i}/N}$. The limit runs through both parities and composite levels.

The original regions, as well as the added ones, must be filled sufficiently long. Their signed powers and the decoration are fixed before N. There is no asserted uniform filling threshold over different diagrams or a simultaneous growing-diagram limit. Preserving an endpoint pairing does not preserve the isotopy type of the filled knot.

Circle patterns and ideal-polyhedral descriptions of fully augmented links are classical; see Purcell [6]. Gluing along the associated thrice-punctured spheres belongs to the setting of belted sums, studied for fully augmented links by Morgan–Ransom–Spyropoulos–Trapp–Ziegler [7]. Volume-conjecture results for Whitehead chains and augmented knotted trivalent graphs appear in van der Veen [8, 9]. We do not claim that the knot families here are disjoint from those classes. The result proved here is the finite, even terminal construction enforcing the specified normalized inequality on an arbitrary fixed marked simple sphere.

2 The imported geometry and coefficient criterion

We state precisely what is used from the companion papers. This allows the local construction below to be read without importing another full asymptotic argument.

Proposition 2.1 (Companion interface)**.** *For a fixed marked matched sphere:*

(i) *Proposition 2.1 and Section 2 of [4] construct the packing, its supporting right-angled polyhedra and the original augmented exterior with labeled disk curves. The complete potential is*

$$P_D(q) = \sum_{\text{triangles } t} \mathcal{Q}((e^{q_f})_{f\in t}) - \sum_{f \text{ ordinary}} q_f.$$

Its minimum is $\Phi_D = \mathrm{Vol}(Y_D)/(2\pi)$, where Y_D is that exterior; the two protected entries are kept exactly zero.

(ii) *Proposition 4.1, Theorem 9.1 and Corollary 9.2 of [1] identify the actual marked spectral source at level $N \geq 3$, with $d = N - 1$, as*

$$S_{D,N}(\boldsymbol{\sigma}) = \left[\prod_{f \text{ ordinary}} L_f^d\right] \prod_e H_{V_N}(L_{x_e}, L_{b_e}, L_{c_e}, L_{y_e}; \sigma_e)\Big|_{L_u=L_v=0}. \tag{2.1}$$

Here u, v are the endpoints of the mark, and

$$V_N(\sigma; z) = \frac{\chi_N(\sigma) - \chi_N(z)}{\sigma - z}, \qquad \chi_N(z) = 2T_N(z/2) + 2$$

is a polynomial divided difference. The local Casimir is substituted for z; $T_N(\cos\phi) = \cos(N\phi)$. The homogeneous polynomial has degree d in the face variables and degree at most d in σ. The theorem retains cups, caps, the original mark, even-level grids and repeated-root normals.

(iii) *Theorem 1.1 of [2] gives the three local upper rates*

$$\begin{aligned} A_{\mathrm{C}} &= X, \\ A_{\mathrm{L}} &= \max\{\log c, \mathcal{Q}(x, b, y) + \tfrac{1}{2}\log c\}, \\ A_{\mathrm{R}} &= \max\{\log b, \mathcal{Q}(x, c, y) + \tfrac{1}{2}\log b\}. \end{aligned} \tag{2.2}$$

The estimates are uniform for all face phases and $\sigma \in [-2,2]$, with fixed positive ordinary moduli and exact protected zeros. An overloaded zero middle removes that one-sided sector. Their hypothesis $x + y > 0$ is (1.4).

(iv) *Definition 3.1 and Theorems 1.1 and 4.3 of [5] give the following implication. A finite shared-weight certificate for every locally allowed noncentral string implies the full volume limit after independently sufficiently long fixed signed fillings. Weak complete balance already supplies such a certificate. The original component test must be a knot.*

For clarity, the certificate in (iv) is a finite list of positive ordinary vectors, one per locally allowed noncentral string ω, with one common strict gap $\gamma > 0$:

$$\sum_{e} A_{e,\omega_e}(L) - \sum_{f \text{ ordinary}} \log L_f \le \Phi_D - \gamma.$$

Each face has a single shared weight within a vector. All locally allowed strings are included, even if their eventual global coefficient vanishes.

The last implication has three separate ingredients. Theorem 1.1 of [1] gives the degree-d full-Hermite filling norm $\prod_e (1 + |n_e|/2)$, and Theorem 10.1 fixes the reduced invariant normalization. Theorem 8.1 and Section 14 of [2] supply the actual central expansion; Theorem 1.1 of [3] evaluates its entire signed filling sum with a nonzero leading coefficient. Finally Theorem 1.1 and Corollary 8.1 of [4] identify its absolute action with the smooth primitive filled volume. These statements include both level parities. The construction below supplies the missing full-sector inequality, rather than inferring it from the central contribution alone.

3 The literal terminal insertion

3.1 The product frame and its two strip substitutions

Label the four half-segments of a turned two-strand tangle h_v in cyclic order by

$$(v_0, v_1, v_2, v_3) = (\text{lower left}, \text{lower right}, \text{upper right}, \text{upper left}).$$

A physical quarter-turn T cyclically shifts the tuple. A lower factor F followed by an upper factor G joins F_2 to G_1 and F_3 to G_0, leaving exposed tuple (F_0, F_1, G_2, G_3). These are empty planar joining strips. The unturned old box is $P_v = T^{-1}h_v$, with exposed tuple (v_3, v_0, v_1, v_2).

Let ξ_1, ξ_2 be the current neighbors of its upper half-segments v_1, v_2. Inserting the turned terminal h_w immediately after P_v makes exactly the replacements

old strip	new strips	
$v_1 - \xi_1$	$v_1 - w_1,$	$w_2 - \xi_1$
$v_2 - \xi_2$	$v_2 - w_0,$	$w_3 - \xi_2.$

(3.1)

Insertion immediately before P_v instead gives

$v_3 - \xi_3$	$v_3 - w_3,$	$w_0 - \xi_3$
$v_0 - \xi_0$	$v_0 - w_2,$	$w_1 - \xi_0.$

(3.2)

Both tables follow by substituting the exposed tuple into the same product rule. The neighbors ξ_i may lie outside the local ball. Thus no decomposition of the rest of the diagram into an ordered tree is assumed.

Lemma 3.1 (The oriented local cut)**.** *The first insertion replaces the oriented white triangle* (x, b, y) *by*

$$(x, b, w), \qquad (b, y, w), \qquad (y, x, w). \tag{3.3}$$

The old matching edge xy *is retained, and* bw *is the new matching edge. The two source factors after insertion are*

$$H_f(x, w, c, y)\, H_g(b, y, x, w). \tag{3.4}$$

They are the coupons in their inherited orientation-preserving frames, with no reflection or change of crossing sign.

Proof. One main-segment vertex in the projection cut is assigned to each paired half-segment, and O_v denotes the vertex on the old crossing-circle semicircle. Its two shaded faces are (O_v, v_1, v_2) and (O_v, v_3, v_0). For (3.1), name

$$C = (v_1, \xi_1), \quad B = (v_2, \xi_2), \quad D = (v_1, w_1), \quad A = (v_2, w_0).$$

After insertion, the outside vertices C, B are represented by $(w_2, \xi_1), (w_3, \xi_2)$. The local attached cell has poles O_v, O_w, equator (A, B, C, D), and shaded triangles

$$(O_v, B, C), \quad (O_v, D, A), \quad (O_w, D, C), \quad (O_w, B, A).$$

Glue its first triangle to the old (O_v, C, B) with opposite orientations. The other three are precisely the new shaded faces specified by the strip table. At each of O_v, B, C, removing the two glued shaded sectors and concatenating the remaining sectors is exactly the planar joining rule in (3.1). The four intervening white triangles therefore extend the three old white regions and introduce one new white region w. Their boundary walks give (3.3). This is a labeled octahedral attachment in each projection copy. The other side uses (3.2) and the other shaded face.

The first two triangles of (3.3), paired along bw, have boundary walk $b \to y \to w \to x \to b$. Their cyclic tuple is (b, y, w, x), so their polynomial argument order is (b, y, x, w). The third triangle, paired with the unchanged (y, c, x) along xy, has cyclic tuple (x, w, y, c). This proves (3.4). □

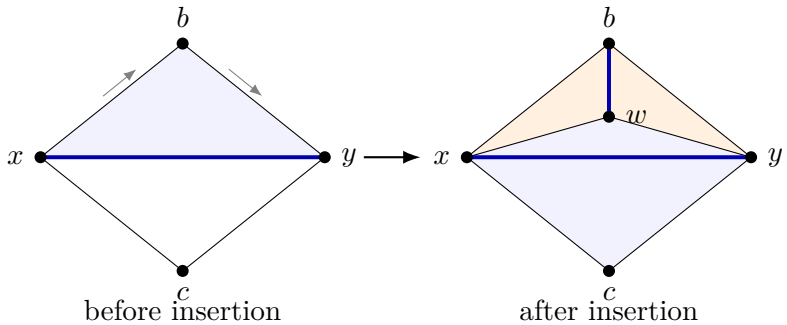


Figure 1: One terminal insertion in the oriented white triangulation. The old pair has cyclic boundary (x, b, y, c). The new vertex is w; the thick matching edges are xy and bw. The retained pair has polynomial order (x, w, c, y); the new pair has order (b, y, x, w). All old boundary edges remain. The arrows record the old first-side orientation.

The operation is a stellar subdivision of one face followed by the displayed matching. The new vertex is fresh and has exactly its three indicated neighbors, so simplicity is preserved. The

three new triangles each belong to exactly one matched pair. A marked boundary edge remains a boundary edge and is never promoted to a matching edge; only bw, which contains a fresh vertex, is added. Consequently the original mark remains valid even when it bounds the subdivided triangle.

3.2 Why an even fan preserves the knot pairing

In the exposed cyclic tuple define the three possible pairings

$$I = \{\{0,3\},\{1,2\}\}, \quad X = \{\{0,2\},\{1,3\}\}, \quad H = \{\{0,1\},\{2,3\}\}.$$

An odd two-strand braid has pairing X, regardless of its sign. Rotating all labels by one preserves X, so a physically turned odd terminal still has this pairing.

In the product of two X's, the first external lower endpoint F_0 follows F_2, G_1, G_3 to the upper-left endpoint. The second follows F_1, F_3, G_0, G_2 to the upper-right endpoint. These are exactly I, and they use every internal joining segment. There is no closed internal component. The through pairing I is an identity on either side of each of I, X, H; this follows by following the same two joining strips. Induction proves:

Lemma 3.2 (Even terminal stacks)**.** *Any even stack of odd turned terminals preserves the four-endpoint pairing of the old tangle, on either side, and creates no internal main component. The signs of the terminal powers are arbitrary.*

This statement concerns strand connectivity. It does not cancel the corresponding braid operators. Every inserted crossing circle remains a distinct component of the augmented link and receives its own filling integer.

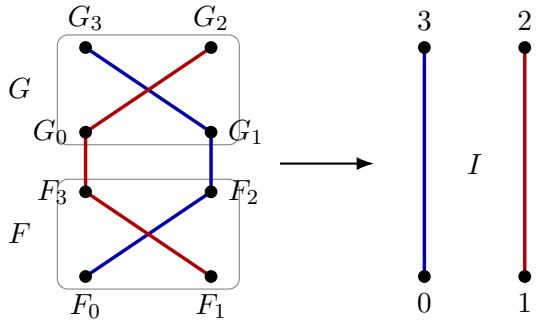


Figure 2: Endpoint paths through two odd turned terminals. The lower factor F is joined to the upper factor G by F_2G_1 and F_3G_0. The blue and red paths connect the external pairs $0-3$ and $1-2$; every internal endpoint lies on one path. The crossings in the drawing represent the pairing X, without fixing the signs of the physical odd powers.

4 A packing extension which fixes every old circle

Lemma 4.1 (The local interstice)**.** *Let the subdivided triangle have curvatures x, b, y, with $x + y > 0$. There is a unique disk in its chosen interstice tangent to those three boundary disks. Its curvature is*

$$b_1 = x + y + b + 2\sqrt{xb + xy + by} = A + b + 2\sqrt{Ab + P}. \tag{4.1}$$

Inserting this disk and replacing the shaded faces gives the complete packing of the subdivided matched sphere, with all previous circles fixed. The assertion includes marked lines.

Proof. First suppose the three boundary curvatures are positive. Write p_i for their centers and r_i for their radii $1/x, 1/b, 1/y$. Subtract the three squared tangency equations $|z - p_i|^2 = (r + r_i)^2$, where $r = 1/k$ is the new radius. Eliminating the two center coordinates gives

$$(k - x - b - y)^2 = 4(xb + xy + by).$$

The disk in the bounded interstice takes the plus sign at equal boundary radii. The two roots never meet for a positive triple, and the bounded gap and its tangent disk vary continuously with the boundary radii. Thus this branch is (4.1) throughout the positive octant.

Here is also a direct model for the labeled supporting faces, which avoids relying on the tangency equation alone. Apply an orientation-preserving Möbius map sending one of the three old contacts to infinity. Rescale so the boundary disks are

$$\Im z \le 0, \qquad \Im z \ge 2, \qquad |z - \mathrm{i}| \le 1.$$

Choose their right-hand interstice. The old shaded outward disk is $\Re z > 0$, and the new white disk is $|z - (2 + \mathrm{i})| < 1$. Its new contacts are $2, 2 + 2\mathrm{i}, 1 + \mathrm{i}$. The replacement shaded outward disks are

$$|z - 1| < 1, \qquad |z - (1 + 2\mathrm{i})| < 1, \qquad \Re z > 2. \tag{4.2}$$

The six old and new contacts form a regular ideal right-angled octahedron. Its three incident white planes agree with the old planes, so the corresponding seams are flat.

To check remote supporting inequalities, use the embedded center triangles supplied by the packing proposition. A bounded shaded contact disk lies inside its center triangle. A nonincident white disk meets that triangle, if at all, through its own incident sectors, whose interiors belong to different center triangles; it therefore misses the shaded disk. Distinct bounded shaded disks also have disjoint interiors. For the omitted outer triangle, all other contacts and bounded disks lie on the inner side of its contact circle and miss its chosen outward disk. These statements persist under the normalization above.

The new white disk and all three shaded disks in (4.2) lie in the old interstice side. They miss every nonincident old outward disk. Conversely old contacts outside the attaching face satisfy the new supporting inequalities. The old and new faces form a closed labeled sphere of exposed supporting facets. The union of the old polyhedron and the octahedron along the attaching triangle is therefore their inward halfspace intersection, with no unintended face or contact. Repeating in the other projection copy realizes precisely Lemma 3.1.

Returning to the original marked coordinate leaves every old circle fixed. One zero curvature is obtained by letting one of the original circles become a line; the tangency formula has the continuous value (4.1). With two zero curvatures the two marked lines are parallel, and the displayed model applies directly: the next disk has the same curvature as the old disk. The case $x = y = 0$ is excluded by (1.4); hence the new curvature is always positive. Every insertion thus extends the complete labeled packing on the original marked sphere. □

The local normalization is used only to verify the face attachment; it is not a new choice of marked radius vector in the source. In the global marked coordinate the protected lines and every old finite circle remain unchanged. The contact model also shows that one insertion increases the complete augmented volume by $2v_{\mathrm{oct}}$, one regular ideal octahedron in each copy. The volume limit below does not require that numerical increment.

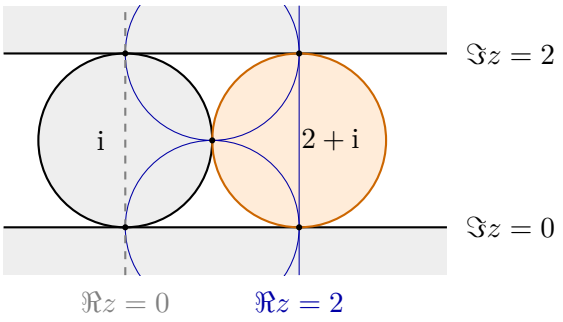


Figure 3: The normalized interstice. The black lines are $\Im z = 0, 2$; the old and new white disks have centers i and $2 + \mathrm{i}$ and radius one. The dashed old shaded boundary is $\Re z = 0$. The blue circles, centered at $1, 1 + 2\mathrm{i}$, and the blue line $\Re z = 2$ bound the three new outward shaded disks. The new disk and those shaded disks are contained in the old right-hand interstice side.

5 The exact fan recurrence and its finite counts

Repeat insertion on one side of a retained original region, always subdividing its current shaded triangle. Let $b_0 = b$. The outer curvatures x, y, and hence A, P, remain fixed. Put $t_j = \sqrt{Ab_j + P}$, taking the nonnegative root. By (4.1),

$$Ab_{j+1} + P = Ab_j + P + A^2 + 2At_j = (t_j + A)^2.$$

Since $A > 0$, this proves the exact solution

$$t_j = t_0 + jA, \qquad b_j = b + 2j\sqrt{Ab + P} + j^2 A. \tag{5.1}$$

There is no approximation in this recurrence, even when $b = 0$. Normalize

$$u = \sqrt{b/A + \rho}, \qquad v = \sqrt{c/A + \rho}.$$

After j, k steps on the two sides, respectively,

$$b_j = A((u + j)^2 - \rho), \qquad c_k = A((v + k)^2 - \rho). \tag{5.2}$$

Choose an integer m nearest $(v - u)/2$, with either choice at a tie, and set

$$\begin{aligned} L &= \max\left\{0, -m, \left\lceil \frac{4 - v}{2} \right\rceil, \left\lceil \frac{4 - u}{2} - m \right\rceil\right\}, \\ j &= 2(L + m), \qquad k = 2L. \end{aligned} \tag{5.3}$$

Both counts are even nonnegative integers and are finite functions of the original curvatures. They satisfy

$$U = u + j \geq 4, \qquad V = v + k \geq 4, \qquad |U - V| \leq 1. \tag{5.4}$$

Indeed the nearest-integer choice gives the last inequality, and the four entries in the maximum give the other conditions. Adding the same nonnegative even number to j, k preserves all three inequalities. This is a finite construction at each region, not an eventual assertion without specified counts.

After each insertion the new crossing region consists of the other two new triangles; neither is subdivided by the next step on the retained region. Hence previously added regions are not changed again. Operations at different original regions occur in different original triangles and preserve their common boundary circles. Their counts can therefore be chosen from the original packing independently, and the finite operations can be performed in any order respecting each side's sequence.

6 The normalized action and strict balance

6.1 A useful elementary comparison

From the definition of e,

$$e(0) = e(1/2) = 0, \qquad e'(a) = \log\cot(\pi a), \qquad e''(a) = -\frac{2\pi}{\sin(2\pi a)}.$$

In particular e is strictly concave and positive in the interior. The function $h(a) = e(a) + a\log(2a)$, extended continuously at zero, vanishes at both endpoints and has

$$h''(a) = -\frac{2\pi}{\sin(2\pi a)} + \frac{1}{a} < 0.$$

Thus $e(a) > -a\log(2a)$ for $0 < a < 1/2$. Choosing $a_i = w_i/(2W)$, $W = \sum_i w_i$, in (1.2) gives

$$\mathcal{Q}(w_1, w_2, w_3) \geq \tfrac{1}{2}\log W, \tag{6.1}$$

strictly if at least two weights are positive. This elementary inequality will settle every newly inserted region.

6.2 An integral which fixes the retained-region comparison

The coarser bound (6.1) alone does not settle the retained region. Its two new middles are large, and their precise relative size matters.

For fixed x, y with $A > 0$ and $b > 0$, the maximizing middle load is

$$\frac{\partial}{\partial \log b}\mathcal{Q}(x, b, y) = \frac{1}{\pi}\arctan\frac{b}{\sqrt{Ab + P}}. \tag{6.2}$$

For two positive entries the tangent ratios are reciprocal. For three entries their sum is $\pi/2$, since $(Ab + P) = (xb + xy + by)$. The Lagrange multiplier equations $e'(a_i) + \log w_i = \text{constant}$ therefore give (6.2), as also recorded in [5, Lemma 4.1].

Homogeneity gives $\mathcal{Q}(x, b, y) - \frac{1}{2}\log b = \mathcal{Q}(x/b, 1, y/b) \to 0$ as $b \to \infty$. To verify the upper limit, a maximizing sequence cannot keep positive load on a disappearing outer weight: its negative logarithm would exceed the bounded continuous entropy. The outer loads therefore tend to zero, and their entropy tends to zero. The vertex allocation at the middle gives the reverse bound. Integrating (6.2) from infinity now proves the absolutely normalized identity

$$\begin{aligned} \mathcal{Q}(x, b, y) &= \tfrac{1}{2}\log b + R(b), \\ R(b) &= \frac{1}{\pi}\int_b^\infty \arctan\frac{\sqrt{At + P}}{t}\,\frac{dt}{t} > 0. \end{aligned} \tag{6.3}$$

The integrand is $O(t^{-3/2})$; the formula also holds when one outer weight vanishes.

Monotonicity of arctangent, $P \geq 0$, and $\arctan z \geq z - z^3/3$ for $z \geq 0$ give

$$R(b) \geq \frac{2}{\pi}\sqrt{\frac{A}{b}} - \frac{2}{9\pi}\left(\frac{A}{b}\right)^{3/2}. \tag{6.4}$$

The arctangent inequality follows by differentiating the difference; its derivative is $z^4/(1+z^2) \geq 0$.

Proposition 6.1 (The retained region)**.** *If U, V satisfy (5.4) and*

$$b' = A(U^2 - \rho), \qquad c' = A(V^2 - \rho),$$

then

$$\mathcal{Q}(x, b', y) + \mathcal{Q}(x, c', y) - \max(\log b', \log c') > \frac{3347}{27720\min(U, V)} > 0. \tag{6.5}$$

Proof. Exchange U, V if necessary so $U \geq V$. The quantities $(U^2 - \rho)^{-1/2}$ and $(V^2 - \rho)^{-1/2}$ are at most $2/\sqrt{63} < \sqrt{3}$, where $2w - 2w^3/9$ is increasing. Apply (6.4) and use $\rho \geq 0$ to obtain

$$R(b') + R(c') \geq \frac{2}{\pi}\left(\frac{1}{U} + \frac{1}{V}\right) - \frac{2}{9\pi}\left(\frac{1}{U^3} + \frac{1}{V^3}\right) \geq \frac{643}{180\pi V}.$$

For the last step, $U \leq V + 1 \leq 5V/4$, $U \geq V$ and $V^{-3} \leq (16V)^{-1}$ give $18/5 - 1/36 = 643/180$. The possible mismatch of the two middles satisfies

$$\tfrac{1}{2}\log(b'/c') = \int_V^U \frac{t}{t^2 - \rho}\, dt \leq \frac{1}{V(1 - 1/64)} = \frac{64}{63V}.$$

Finally $\pi < 22/7$ gives

$$\frac{643}{180\pi} - \frac{64}{63} > \frac{643 \cdot 7}{180 \cdot 22} - \frac{64}{63} = \frac{3347}{27720}.$$

One elementary verification of the strict rational bound is $\frac{22}{7} - \pi = \int_0^1 t^4(1-t)^4/(1+t^2)\, dt > 0$. Subtract the logarithmic mismatch from the two positive remainders in (6.3). This proves (6.5). □

Proposition 6.2 (Every added region)**.** *Every newly inserted region is strictly balanced when it is created, including all protected-zero cases.*

Proof. Its outer pair is (b, b_1), and its middle pair is (y, x) in the actual argument order. Put $A' = b + b_1$. Equation (4.1) gives $A' \geq x + y = A > 0$. By (6.1),

$$\mathcal{Q}(b, x, b_1) + \mathcal{Q}(b, y, b_1) \geq \tfrac{1}{2}\log((A' + x)(A' + y)).$$

Let $m = \max(x, y) > 0$. Since $A' \geq m$, the product on the right is at least $2m^2$. The sum is consequently larger than $\log m$, and hence larger than the logarithm of every positive middle. A zero middle has no corresponding one-sided sector. The proof permits $b = 0$; $b_1 > 0$ and $A > 0$ still hold. □

Proof of Theorem 1.2. For every original region choose j, k by (5.3) and perform the two finite sequences. Lemmas 3.1 and 4.1 preserve the simple matched sphere, its mark and every old circle at each step. Formula (5.1) computes its retained middle curvatures exactly. Proposition 6.1 gives strict balance at that region after its two sequences. Proposition 6.2 settles every added region, which no later step changes. Different original regions keep their common old curvatures, so the argument applies to all of them on one marked packing. The total number of steps is a finite sum of the explicit counts. Finally Lemma 3.2 gives the endpoint and component assertions because both counts at every region are even. □

7 The actual refined source and the filled limit

The sphere changes, so its invariant also changes. It is important to identify the new source rather than substitute a scalar multiplication rule for terminal tangles. Lemma 3.1 fixes the exact two new argument orders in (3.4). Every other local frame is untouched. One new vertex and one new matched pair are introduced. In (2.1), every original ordinary face keeps its target $d = N - 1$; the new ordinary face receives one target d. No additional coefficient is taken at the protected faces.

These statements can also be seen directly in the local winding data of the exact diagram theorem. Let M_f count middle incidences and T_f count outside minus inside Morse turns, temporarily restoring the cap at the mark. The local change is

$$\Delta M_b = -1, \quad \Delta M_x = \Delta M_y = 1, \quad M_w = 1, \qquad \Delta M_c = 0.$$

For a bounded face, $M_f - T_f = 2$; for the exterior face the right side is -2. Therefore

$$\Delta T_b = -1, \quad \Delta T_x = \Delta T_y = 1, \quad T_w = -1, \qquad \Delta T_c = 0.$$

The old target exponents are unchanged, and the new exponent $\frac{d}{2}(T_w - M_w) = -d$ supplies exactly the new target. At even N, the face parity changes by one across a strand edge and by zero across a matching edge. Because xy is matching, $\tau_x = \tau_y$, and the unique new clock is $\tau_w = \tau_b = \tau_x + 1$ modulo two. Thus the new matching edge bw and all the new strand edges have the prescribed parities. These computations agree with the full source theorem, including its unchanged reduced mark. The extra coupon factor N^{-1} is canceled by the new primitive N-point face grid; there is no residual quantum-dimension factor.

Lemma 7.1 (A certificate at the complete weights)**.** *For a fixed strictly balanced marked sphere, its complete packing itself is an unabridged shared-weight certificate.*

Proof. For an allowed left label with positive overloaded middle c,

$$X - A_{\mathrm{L}} = \min\{X - \log c,\ \mathcal{Q}(x, c, y) - \tfrac{1}{2}\log c\} > 0.$$

The first entry is strict balance. The second is the positive remainder in (6.3), since $x + y > 0$. The right label is identical with b, c exchanged. There are finitely many allowed sides. If there are any, let γ be the minimum of their positive losses. At the complete vector, the all-central shared action is Φ_D. A noncentral string subtracts at least one loss and no negative loss, so its full shared rate is at most $\Phi_D - \gamma$. This includes globally vanishing strings as well. If no one-sided label is allowed the certificate is vacuous. □

Proof of Corollary 1.3. Theorem 1.2 produces a fixed marked simple sphere, and Lemma 7.1 gives its unabridged certificate. Apply [5, Theorem 1.1] through Proposition 2.1. Every original and added crossing circle retains its own primitive disk slope. Choose $\epsilon_e = \pm 1$ at added circles, so (1.1) is odd for every integer filling coefficient. The old parity pattern and Lemma 3.2 make the final main component a knot for every such choice.

The full complementary source has a fixed strict exponential gap. The Hermite norm costs only the constant $\prod_e (1 + |n_e|/2)$ for this fixed filling vector. The nonzero central action tends to the complete action as all $|n_e|$ grow independently. Enlarge the threshold so its rate exceeds the complementary upper rate and so the primitive geometric completion theorem applies. The full

invariant then has the same rate as that nonzero contribution. The original framing factor has modulus one, and the primitive action-volume identity gives (1.5). Taking $|p_e|$ sufficiently large implies the required $|n_e|$ bound uniformly in the finitely many retained ϵ_e's. No restriction on ratios of the powers is used. □

There is a combinatorial supply of core knot patterns if none is specified initially. Form the graph whose vertices are the untwisted main components and whose edges are the crossing regions. It is connected because the original contracted four-valent graph is connected. Choose a spanning tree and switch exactly those boxes to odd pairing. After any subset of these switches, the current components are the components of that forest. The next tree edge joins two different components and merges them. Induction gives one knot. The signs of these odd powers are arbitrary, and all subsequent full twists preserve the pairing. Thus the theorem can start from the finite sphere and matching, with the parity pattern chosen explicitly.

8 Exact finite reproduction

The uniform proofs are the recurrence, the integral inequalities, the local cut and the finite induction above. The companion checker provides a second way to catch errors in their finite formulas:

```
python3 papers/terminal-decorations/check.py --output /tmp/terminal-check
```

It uses the standard Python library and the existing exact terminal-fan verifier. Polynomial identities are compared in the ring with relation $t^2 = Ab + P$. The rational constant 3347/27720 is checked exactly. Oriented face cycles are used to recover the new quartet, and connected components of the literal pairing graph test both terminal products. The runner also checks the matching, the protected edge and the even-level parity extension through sixteen recorded insertions. Its forty-nine rational count examples are checks of the formula (5.3); the all-real proof is the nearest-integer argument above.

The output records source hashes and the exact scope of these checks. It does not evaluate a knot invariant or replace the all-length packing, analytic asymptotic or metric-completion proofs. The new estimates in this paper require no numerical packing optimizer or sampled saddle point.

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